



DIGITAL LOCK PROJECT REPORT

CSE211S: LOGIC DESIGN

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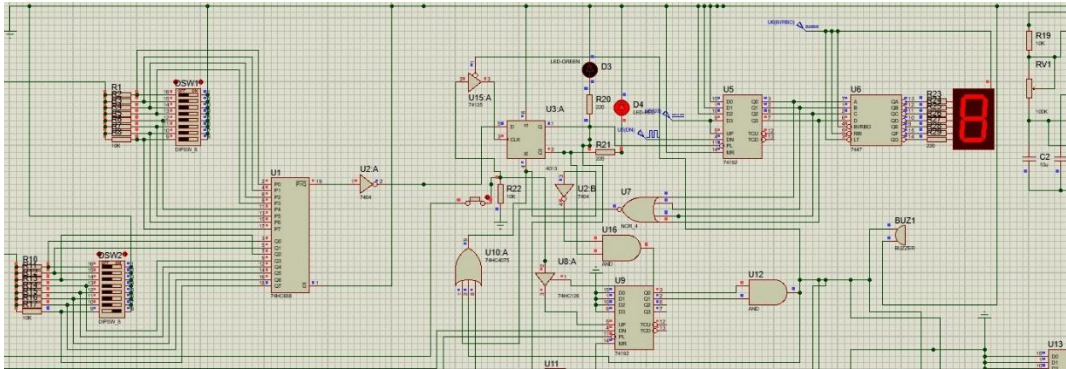
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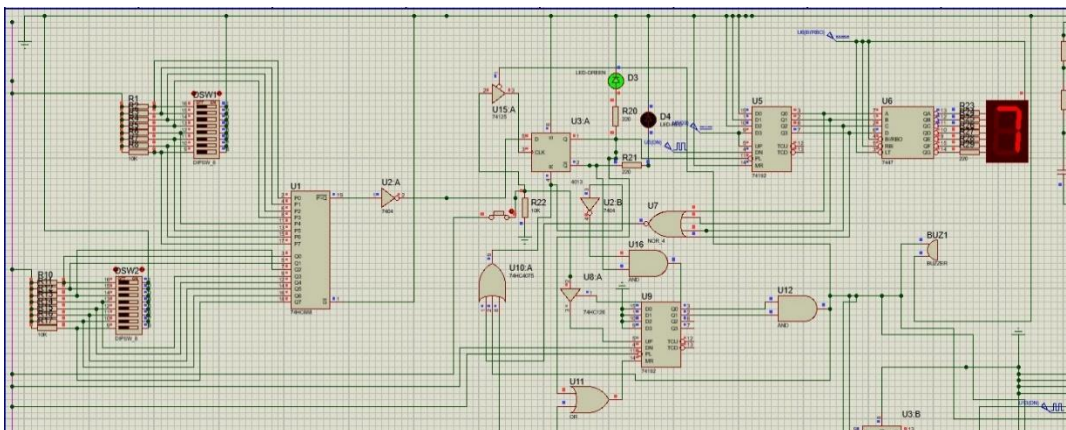
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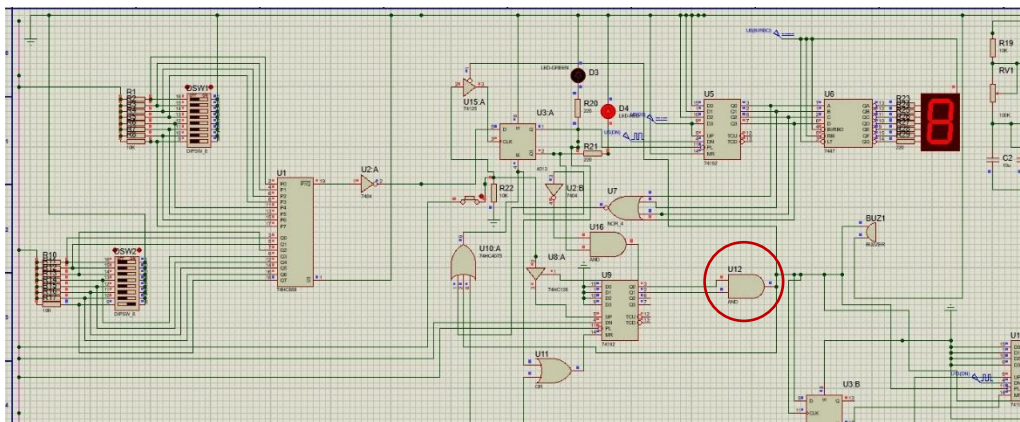
SIMULATION



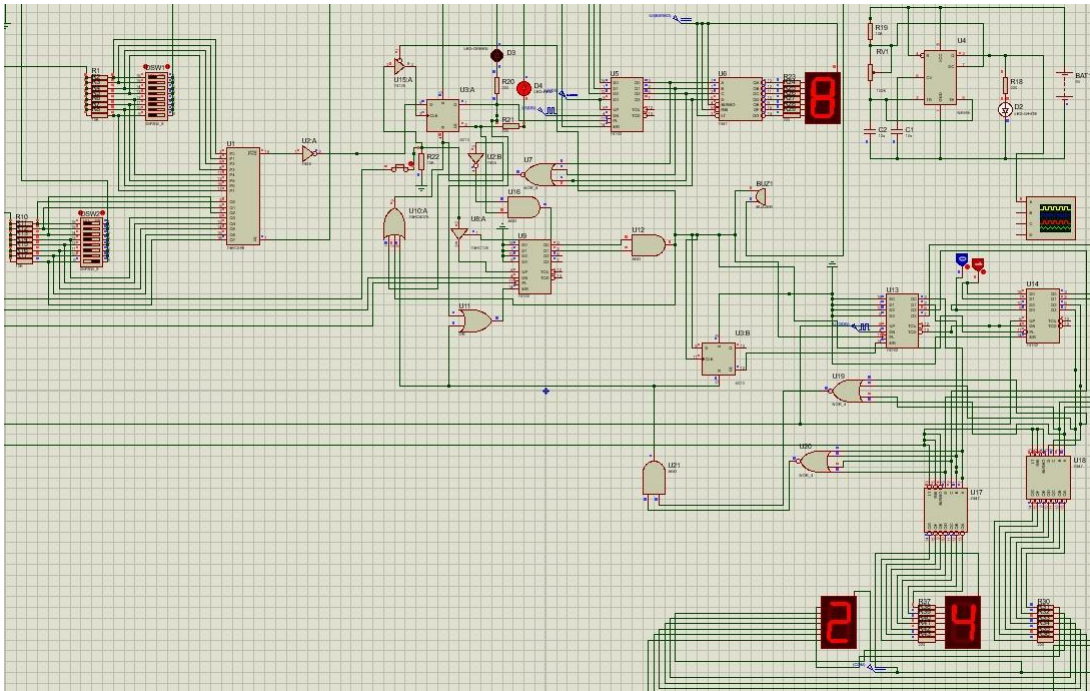
Here, the code is wrong and thus the system is at the '0' state (The red LED is on and the counter is loaded with the number '8')



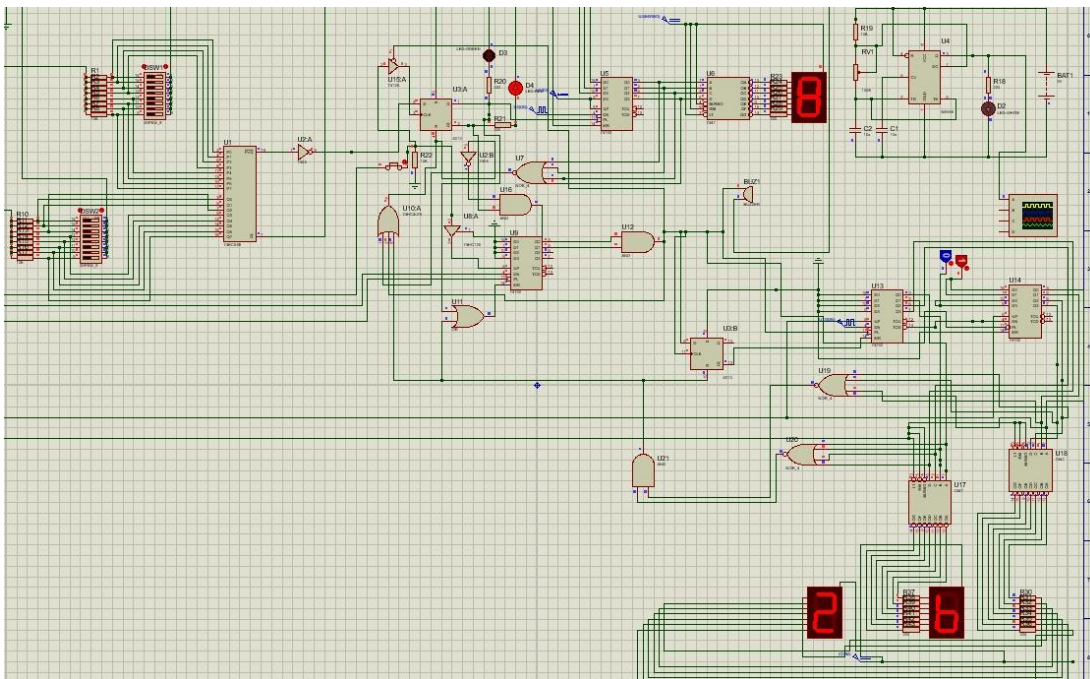
Now, the code is right and thus the system turned to the '1' state (The green LED is on and the counter started its down counting)



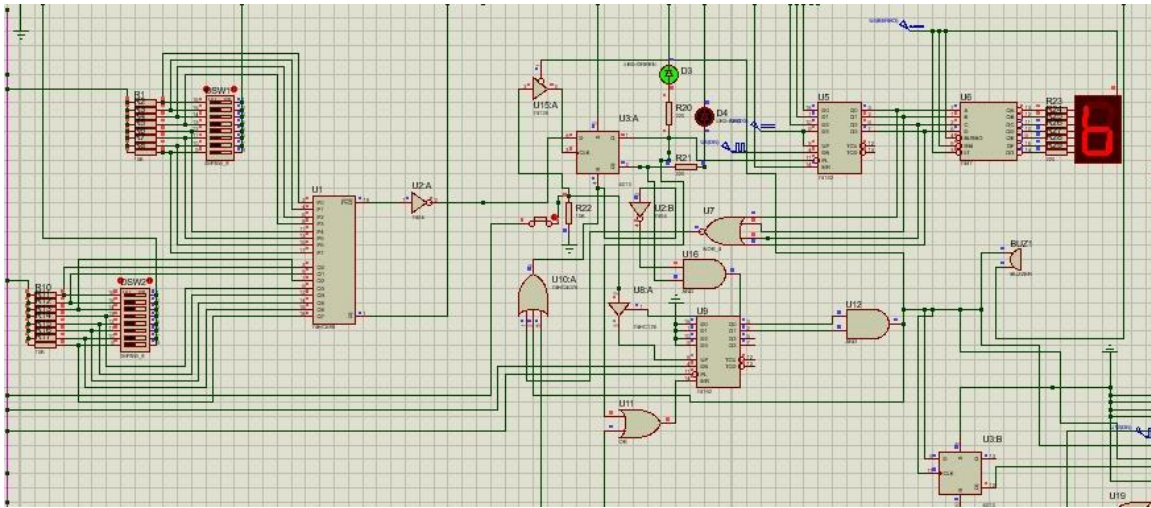
One wrong input has been inserted and so you can see how the AND gate from the counter's outputs indicates the number '1'.



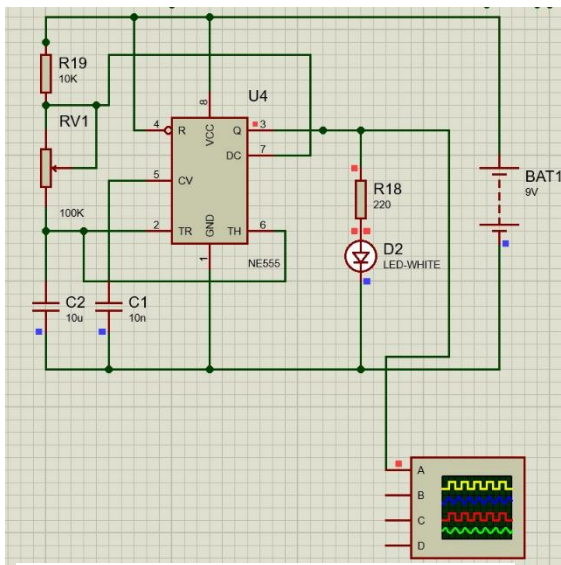
Three wrong inputs have been inserted and thus the buzzer is on and the 30 seconds counter started its down counting.



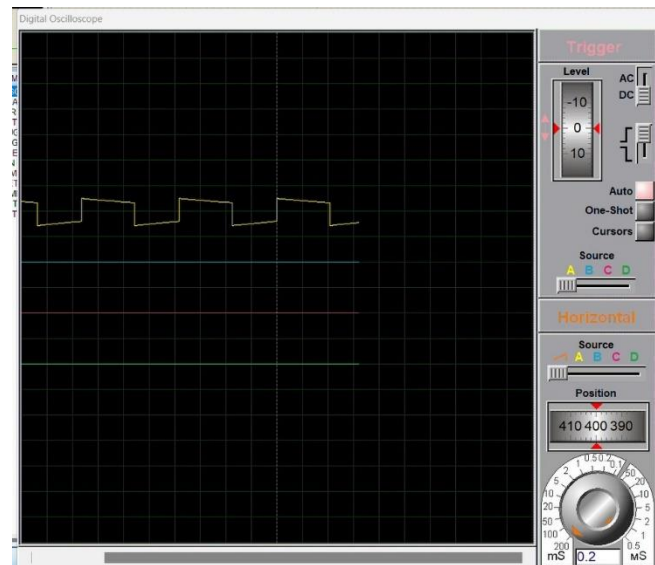
During the 30 seconds, even if the user inserted the right code, the system remains disabled (remains in the '0' state).



After the 30-second timeout expires, the system returns to normal input-receiving mode. In this case, it received the correct code and transitioned to its '1' state.



The 555 timer circuit that generates a 1 Hz clock signal that drives and synchronizes the system's counters.



The 555 timer circuit waveform.



TOOLS AND COMPONENTS

Tool/ Component	Cost (EGP)
5V DC/ 2A Power Adaptor	50.00
Breadboards	140.00
Connecting Jumper Wires	100.00
DIP Switches 8 Way	20.00
Resistor Networks for DIP Switches	8.00
IC 74688 - 8-bit Magnitude Comparator	50.00
IC 7474 - Dual D-type Flip Flop	15.00
Push Button	1.50
NE555P - Timer IC	2.00
Potentiometer 100K Ω	6.00
Resistance 10K Ω	5.00
Resistance 220 Ω	7.00
Capacitor 10uF	0.75
Ceramic Capacitor 10nF	0.25
LED 5mm Red Color	0.50
LED 5mm Green Color	0.50
IC 74192 - Synchronous Up/Down 4-bit Counter	60.00
IC 7447 - BCD to 7-Segment Decoder/Driver	120.00
7 Segment Display	20.00
Buzzer	8.00
IC 7404 – NOT gate	15.00
IC 4072 - Dual 4 input OR Gate	30.00
IC 7408 - Quad 2 input AND Gates	15.00
IC 4071 - Quad 2 input OR Gates	20.00

TROUBLESHOOTING AND TECHNICAL CHALLENGES

During the development and testing phases of our circuit, we encountered several technical hurdles ranging from software limitations in Proteus to complex hardware power management issues. Each problem required a systematic approach to diagnose and solve, involving deep research and peer consultation.

1. Navigating Simulation Constraints

The transition from a theoretical design to a digital model in Proteus was our first hurdle. We discovered that simulation software, while powerful, is only as good as its component libraries. Several specialized Integrated Circuits (ICs) we had selected lacked active models in the Proteus database. This forced us to go back to the drawing board and pivot our design toward ICs that were not only available in local shops but also had verified simulation libraries. This ensured that our logic was "software-proven" before we committed to physical assembly.



2. Overcoming Floating Inputs with Pull-Down Resistors

One of the most persistent issues we faced involved the Clock (CLK) input of our D Flip-Flop. We had connected a push button to trigger the clock, but we encountered significant "undefined behavior" whenever the button was not being pressed.

- **The Problem (The Floating State):** In digital electronics, an input pin must always have a clear path to either voltage (High) or ground (Low). When the push button was open, the circuit was physically disconnected. This created a "floating" node—essentially acting like a small antenna that picked up ambient electromagnetic noise. This noise caused the Flip-Flop to perceive rapid, random clock pulses, leading to erratic and unpredictable outputs.
- **The Simulation Snag:** While we initially considered using logic gates to "clean" the signal, the Proteus simulation environment struggled to model the transition correctly. This forced us to go back to fundamental circuit design to find a hardware-reliable solution.
- **The Solution (Pull-Down Resistor):** We integrated a pull-down resistor between the CLK pin and the ground.
 - **How it works:** When the button is not pressed, the resistor "pulls" the input pin down to 0V (Ground), ensuring a rock-solid Logic Low state and draining away any stray noise.
 - **When pressed:** When the user pushes the button, the direct connection to the power supply overcomes the resistor's path, bringing the pin to a Logic High.

This simple but effective addition eliminated the "noise-induced" errors, providing us with a clean, intentional trigger every time the button was pressed.

3. Resolving Unexpected Display Flickering

In our initial setup, we overlooked the necessity of current-limiting resistors between the decoder and the 7-segment display. This led to erratic behavior; the displays would glitch or show ghosting segments. We realized that without these resistors, the display segments were pulling inconsistent amounts of current, creating "noise" in the decoder's output. By calculating and inserting the correct resistance for each segment, we stabilized the visual output and ensured the longevity of the display LEDs.

4. Diagnosing "Hidden" Hardware Fatigue

After weeks of testing, we encountered a mysterious failure: the timer stopped toggling entirely, and the output would "flame" (stay constantly high) as soon as power was applied. Structurally, our connections were perfect, which made the error incredibly frustrating. After seeking mentorship from older peers, we learned about jumper wire fatigue. We discovered that the jumper connecting Pin 2 (Trigger) and Pin 6 (Threshold) had lost its internal conductivity. Because these two pins are responsible for sensing voltage



levels and triggering the internal transistors, the faulty wire had trapped the timer in a permanent high state. Replacing this single jumper restored the entire circuit to life.

5. Improper Breadboard Wiring

Another challenge was related to improper jumper connections on the breadboard. In some cases, loose jumper wiring caused incomplete clock or carry/borrow connections between counter stages. As a result, the counter exhibited an abnormal behavior, such as counting only even numbers or only odd numbers, instead of a continuous sequence. This problem was resolved by ensuring that all jumper wires were firmly seated or replacing the jumpers.

6. Simulation-Hardware Behavior Mismatch

One of the main challenges during the hardware implementation was dealing with floating pins. The circuit worked correctly in the simulation, while the breadboard version showed incorrect behavior due to unconnected control pins. Floating inputs caused noise-sensitive signals, leading to unexpected counting, skipping numbers, and false triggering. This problem was resolved by carefully reviewing each IC datasheet and ensuring that all unused or control pins were properly connected to either VCC or GND. This highlighted the importance of considering real hardware behavior rather than relying solely on simulation results.